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Saveetha Institute of Medical And Technical Sciences

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Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT

Deployment Models, Business Benefits, Challenges, and Practical Applications

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Introduction

- Multi-Cloud: Using services from two or more cloud providers, such as AWS, Azure, and Google Cloud.
- Hybrid Cloud: Combining a private cloud with a public cloud to store and manage data.
- Business Benefits: These strategies reduce costs, improve flexibility, and increase performance.
- Challenges: They may face security risks, complex management, and data integration problems.
- Practical Examples: Companies use multi-cloud and hybrid cloud for data storage, backup, application hosting, and disaster recovery.

Objectives

01

Define the models

Clarify public, private, hybrid and multi-cloud deployment options.

02

Design the architecture

Show how control, network, identity and data layers fit together.

03

Weigh benefits and challenges

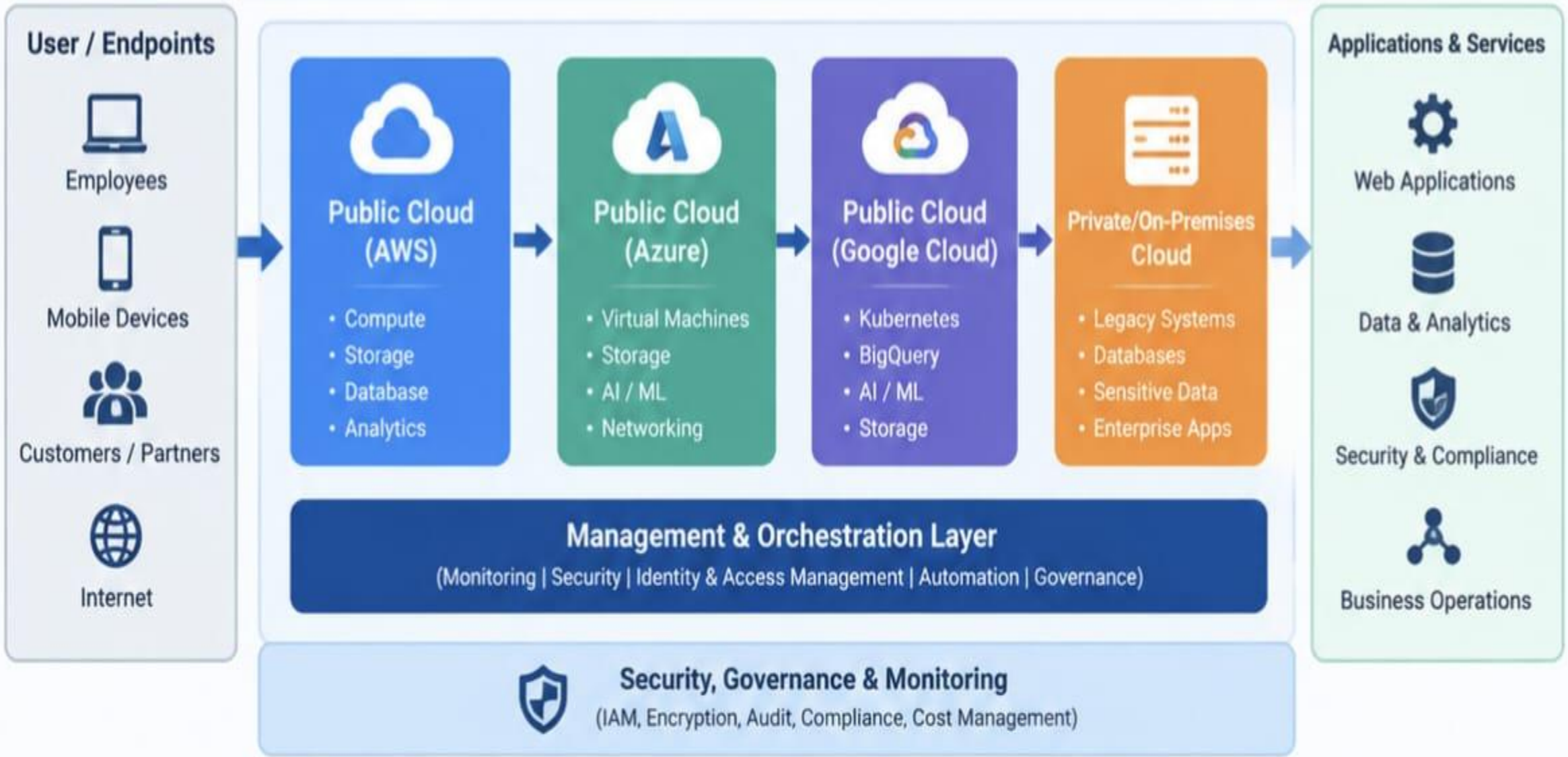
Assess business value against cost, complexity and risk.

04

Learn from practice

Apply the findings to realistic enterprise scenarios.

Cloud Architecture / System Architecture



Hybrid vs. Multi-Cloud

Aspect	Hybrid cloud	Multi-cloud
Core idea	Private or on-prem and public cloud run as one integrated environment	Services from two or more public providers used together
Main driver	Control of sensitive data, legacy reuse, gradual migration	Resilience, best-of-breed services, negotiating leverage
Typical architecture	Data center plus public cloud linked by dedicated connectivity	Workloads or tiers spread across providers behind a common control layer
Key strength	Data stays local while public cloud adds elasticity	No single-provider dependency; best service for each task
Main challenge	Integrating two very different worlds; latency between them	Skills, tooling and security consistency across providers
Best fit	Regulated industries and large legacy estates	Global, digital-native or risk-sensitive enterprises

Practical Examples

Illustrative enterprise scenarios



Banking and finance

HYBRID

Core ledgers and regulated customer data stay in private data centers; analytics, fraud models and mobile front-ends scale in public cloud.



Healthcare

HYBRID

Patient records remain on controlled infrastructure while imaging AI and research workloads use elastic cloud compute.



Retail and e-commerce

MULTI-CLOUD

Peak-season traffic bursts across providers, with CDN and analytics chosen per region for speed and cost.



Manufacturing

HYBRID + EDGE

Plant-floor systems run at the edge for low latency, with telemetry sent to the cloud for predictive maintenance.

Enabling platforms AWS Outposts · Azure Arc · Google Distributed Cloud · Red Hat OpenShift · VMware Cloud · Terraform

Conclusion

1

Hybrid and multi-cloud are complementary

Most enterprises end up using both, guided by workload needs.

2

Architecture matters more than vendor choice

A common control, network, identity and data layer is the foundation.

3

Benefits come with real complexity

Resilience and flexibility need skills, governance and FinOps discipline.

4

Start with workloads, not platforms

Place each workload where its data, risk and cost profile fit best.